

RiskLens Project Report

Early Detection, Better Outcomes

1. Introduction

RiskLens is an AI-powered student risk assessment platform designed to help educators identify at-risk students early and provide students with a clear understanding of their academic standing. By combining advanced machine learning with Google Gemini's generative AI, RiskLens transforms raw academic data into actionable intervention strategies.

2. Step-by-Step Workflow

1. Secure Authentication

Users (Teachers or Students) sign up/in via Supabase, which manages role-based access to specialized dashboards.

2. Data Ingestion

Teachers upload CSV files for batch processing, while students can manually enter their own study metrics.

3. Automated Realignment

A fuzzy matching algorithm automatically maps varied CSV column names (e.g., 'Study Time' vs 'Hours_Studied') to the system model.

4. ML Prediction

A Scikit-Learn pipeline classifies students as 'Low Risk' or 'High Risk' and calculates a precise risk probability percentage.

5. AI Insight Generation

The Google Gemini AI analyzes the risk patterns to generate human-readable reports and specific, actionable interventions.

6. Outcomes Display

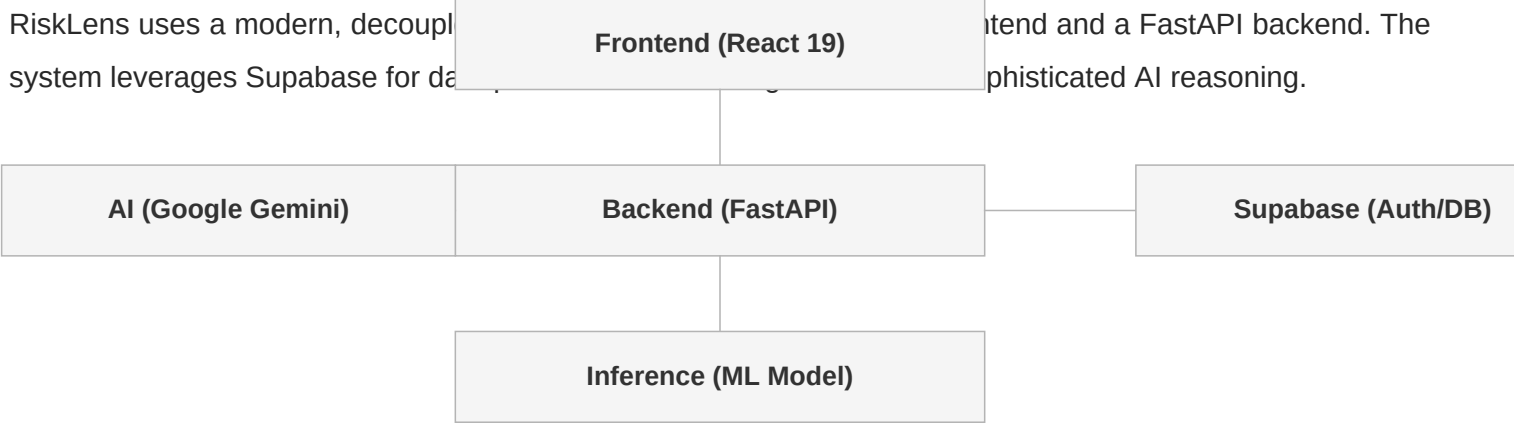
Teachers receive comprehensive class-wide analytics, while students get personalized 'Risk Gauges' with encouraging tips.

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3. Project Architecture

RiskLens uses a modern, decoupled frontend and a FastAPI backend. The system leverages Supabase for data storage and sophisticated AI reasoning.



Key Components:

- Client: React, Framer Motion, Recharts
- API Server: FastAPI, Uvicorn, Pandas
- Intelligence: Scikit-learn, RapidFuzz, Google Gemini Pro
- Infrastructure: Supabase Auth & PostgreSQL